

Applied external wrench and CRG input — tmp

World/LWA components; wrist-origin equivalent torque includes the COM→body-origin shift; solid: applied, dashed: CRG; causally aligned applied→CRG RMSE force/torque = 1.26e-14 N/1.75e-16 N·m (delay 2 ms)

— applied x — applied y — applied z - - - CRG input x - - - CRG input y - - - CRG input z

